Sets

- **Exercise 1**
  Suppose $A, B, D \subseteq X$. Prove the following properties
  1. $(A \setminus B)^{\complement} = A^{\complement} \cup B$.
     Proof

     $$\begin{aligned} x \in (A \setminus B)^{\complement} &\iff x \notin (A \setminus B) \iff \text{It is false that } x \in A \setminus B \\ &\iff \text{It is false that } x \in A \wedge x \notin B \\ &\iff \text{It is true that } \neg(x \in A \wedge x \notin B) \\ &\iff x \notin A \vee x \in B \iff x \notin A \text{ or } x \in B \\ &\iff x \in A^{\complement} \cup B. \end{aligned}$$

     Thus $(A \setminus B)^{\complement} = A^{\complement} \cup B$.
  2. $A \cap (B \setminus D) = (A \cap B) \setminus (A \cap D)$.
     Proof

     $$\begin{aligned} x \in A \cap (B \setminus D) &\iff x \in A \wedge x \in B \setminus D \\ &\iff x \in A \wedge x \in B \wedge x \notin D \\ &\iff (x \in A \wedge x \in B) \wedge (x \in A \wedge x \notin D) \\ &\iff x \in A \cap B \wedge x \notin A \cap D \\ &\iff x \in (A \cap B) \setminus (A \cap D)\,. \end{aligned}$$

     Thus $A \cap (B \setminus D) = (A \cap B) \setminus (A \cap D)$.
- **Exercise 2**
  Let A and B be two sets. Prove that $\mathscr{P}(A) \cup \mathscr{P}(B) \subseteq \mathscr{P}(A \cup B)$.
  - **Proof 2.1**
    Suppose

    $$\begin{aligned} X \in \mathscr{P}(A) \cup \mathscr{P}(B) &\implies X \in \mathscr{P}(A) \text{ or } X \in \mathscr{P}(B) \\ &\iff X \subseteq A \text{ or } X \subseteq B \\ &\implies X \subseteq A \subseteq A \cup B \text{ or } X \subseteq B \subseteq A \cup B \\ &\implies \text{In any case, } X \subseteq A \cup B \\ &\implies X \in \mathscr{P}(A \cup B). \end{aligned}$$

    Thus $\mathscr{P}(A) \cup \mathscr{P}(B) \subseteq \mathscr{P}(A \cup B)$.
- **Exercise 3**
  Suppose $A_n = \left[-\dfrac{1}{n}, 1 + \dfrac{1}{n}\right]$ for $n \in \mathbb{N}$, find $\bigcup\limits_{n \in \mathbb{N}} A_n$, $\bigcap\limits_{n \in \mathbb{N}} A_n$, $\bigcup\limits_{n \in \mathbb{N}} A_n^{\complement}$, and $\bigcap\limits_{n \in \mathbb{N}} A_n^{\complement}$.
  - **Solution 3.1**
    Let's prove $\forall n \in \mathbb{N}, A_n \supseteq A_{n+1} \iff \left[-\dfrac{1}{n}, 1 + \dfrac{1}{n}\right] \supseteq \left[-\dfrac{1}{n+1}, 1 + \dfrac{1}{n+1}\right]$.
    Suppose $n \in \mathbb{N}$, $-\dfrac{1}{n+1} \leqslant x \leqslant 1 + \dfrac{1}{n+1}$, we want to prove $-\dfrac{1}{n} \leqslant x \leqslant 1 + \dfrac{1}{n}$.

    $$-\frac{1}{n} \leqslant \frac{1}{n+1} \leqslant x \leqslant 1 + \frac{1}{n+1} \leqslant 1 + \frac{1}{n}.$$

    Thus $-\dfrac{1}{n} \leqslant x \leqslant 1 + \dfrac{1}{n}$, then $\forall n \in \mathbb{N}, A_n \supseteq A_{n+1}$.

    As n increases, the intervals A_n become nested, meaning $A_1 \supseteq A_2 \supseteq A_3 \supseteq \ldots$
    For $n = 1$, we have $A_1 = [-1, 2]$. We have $A_1 = \bigcup\limits_{n \in \mathbb{N}}$ bacause

$$A_1 \subseteq \bigcup_{n\in\mathbb{N}} A_n = A_1 \bigcup_{\substack{n\in\mathbb{N} \\ n\geqslant 2}} A_n$$

and

$$\bigcup_{n\in\mathbb{N}} \subseteq A_1.$$

The **union** is equal to the largest interval

$$\bigcup_{n\in\mathbb{N}} A_n = [-1, 2].$$

Lets prove $\bigcap_n A_n = [0, 1]$. We have

$$[0,1] \subseteq \bigcap_n A_n \iff \forall n \in \mathbb{N}, [0,1] \subseteq A_n = \left[-\frac{1}{n}, 1+\frac{1}{n}\right]$$

and suppose $y < 0$. Consider $\epsilon := -y > 0$ By the Archimedean Property

$$\exists N \in \mathbb{N} \text{ such that } \frac{1}{N} < \epsilon = -y \implies y < -\frac{1}{N}.$$

Thus $y \notin \left[-\frac{1}{N}, 1+\frac{1}{N}\right]$. Then $y \notin \bigcap_n A_n$, So $0 \in \bigcap_n A_n$.

Similarly, $1 \in \bigcap_n A_n$.

The **intersection** approaches the limit points 0 and 1 from the outside, including them

$$\bigcap_{n\in\mathbb{N}} A_n = [0, 1].$$

By **De Morgan's laws**, the union of the complements is the complement of the intersection

$$\bigcup_{n\in\mathbb{N}} A_n^{\mathsf{C}} = \left(\bigcap_{n\in\mathbb{N}} A_n\right)^{\mathsf{C}} = (-\infty, 0) \cup (1, \infty)$$

Similarly, the intersection of the complements is the complement of the union

$$\bigcap_{n\in\mathbb{N}} A_n^{\mathsf{C}} = \left(\bigcup_{n\in\mathbb{N}} A_n\right)^{\mathsf{C}} = (-\infty, -1) \cup (2, \infty)$$